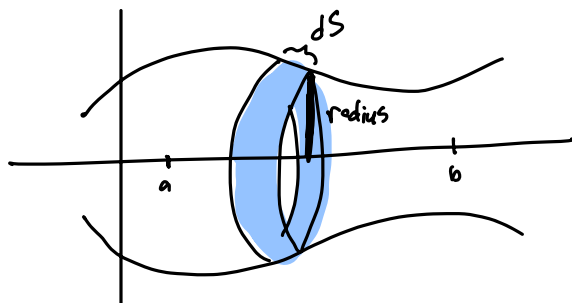
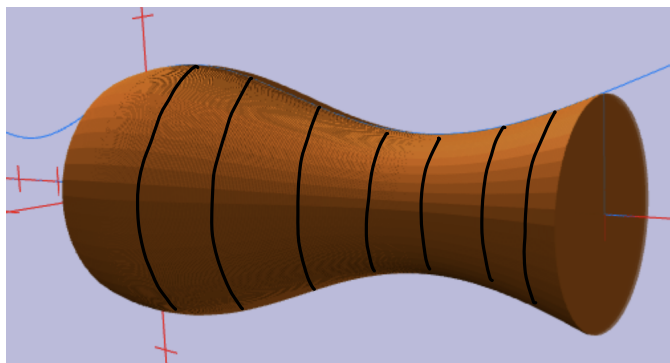


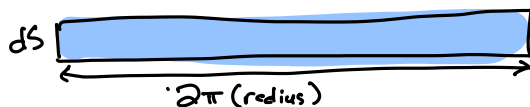
8.2 Area of a Surface of Revolution

Plan: Take a portion of a curve, rotate it around an axis, and find the area of the surface that is generated.



Band Area

Length = circumference

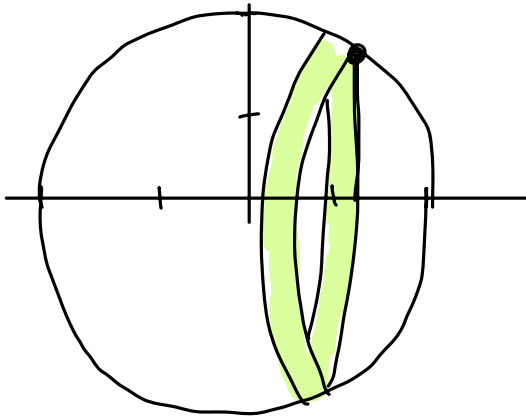


$$\text{Area of Band} = 2\pi(\text{radius})ds$$

$$\text{Surface area of Solid} = 2\pi \int_a^b (\text{radius}) ds$$

0.0.1 Example:

Find the surface area of the sphere formed by rotating the curve $y = \sqrt{4-x^2}$, from $x = -2$ to $x = 2$ around the x -axis.



$$S.A. = 2\pi \int_{-2}^2 \text{radius } dS$$

$$\text{radius} = y = \sqrt{4-x^2}$$

Find dS :

$$\left(\frac{dy}{dx}\right)^2 = \left(\frac{-2x}{2\sqrt{4-x^2}}\right)^2 = \left(\frac{-x}{\sqrt{4-x^2}}\right)^2 = \frac{x^2}{4-x^2}$$

$$dS = \sqrt{1 + \frac{x^2}{4-x^2}} dx$$

$$= \sqrt{\frac{4-x^2}{4-x^2} + \frac{x^2}{4-x^2}} dx$$

$$= \sqrt{\frac{4}{4-x^2}} dx$$

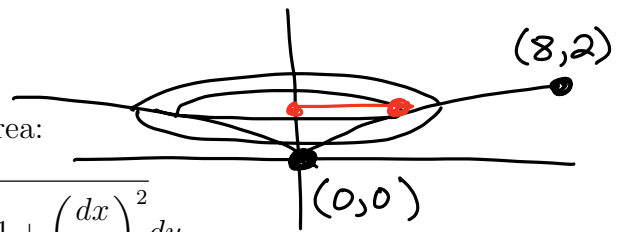
$$S.A. = 2\pi \int_{-2}^2 \cancel{\sqrt{4-x^2}} \frac{\sqrt{4}}{\cancel{\sqrt{4-x^2}}} dx = 2\pi \int_{-2}^2 2 dx = 4\pi \int_{-2}^2 dx$$

$$= 4\pi x \Big|_{-2}^2 = 4\pi [2 - (-2)] = 4\pi (4) = \boxed{16\pi}$$

Note: You can use either form of dS when finding surface area:

$$dS = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \quad \text{or} \quad dS = \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

radius = x



0.0.2 Example:

Find the area of the surface formed by rotating the curve $y = \sqrt[3]{x}$, from $x = 0$ to $x = 8$ around the y -axis. Do it both ways.

with dx

$$y = x^{1/3}$$

$$\frac{dy}{dx} = \frac{1}{3} x^{-2/3} = \frac{1}{3x^{2/3}}$$

$$dS = \sqrt{1 + \frac{1}{9x^{4/3}}} dx$$

radius = x

$$S.A. = 2\pi \int_0^8 x \sqrt{1 + \frac{1}{9x^{4/3}}} dx$$

$$u = 1 + 9y^4$$

$$du = 36y^3 dy$$

$$\frac{1}{36} du = y^3 dy$$

$$\text{Top: } u = 1 + 9(2)^4 = 145$$

$$\text{Bottom: } u = 1 + 9(0)^4$$

with dy

$$y^3 = x$$

$$(3y^2)^2 = \left(\frac{dx}{dy}\right)^2 = 9y^4$$

$$dS = \sqrt{1 + 9y^4} dy$$

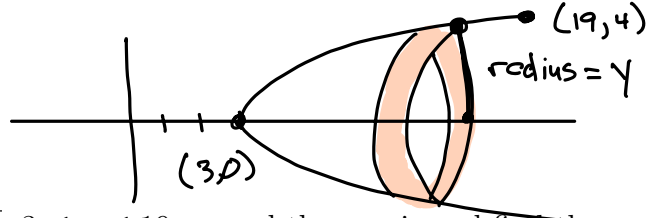
radius = $x = y^3$

$$S.A. = 2\pi \int_0^2 y^3 \sqrt{1 + 9y^4} dy$$

$$= 2\pi \int_1^{145} \sqrt{u} \frac{du}{36}$$

$$\left[\frac{2\pi}{36} \cdot \frac{2}{3} u^{3/2} \right]_1^{145} = \frac{\pi}{27} (145^{3/2} - 1)$$

$$\approx \boxed{203.044}$$



0.0.3 Example:

Rotate the curve $y = \sqrt{x-3}$, $3 \leq x \leq 19$ around the x -axis and find the surface area.

$$y^2 = x - 3$$

$$y^2 + 3 = x \leftarrow \text{use this + } dy$$

$$x = y^2 + 3$$

$$\left(\frac{dx}{dy}\right)^2 = (2y)^2 = 4y^2$$

$$dS = \sqrt{1 + 4y^2}$$

$$S.A. = 2\pi \int_0^4 y \sqrt{1 + 4y^2} dy$$

$$\boxed{\begin{array}{l} u = 1 + 4y^2 \\ du = 8y dy \end{array}} = 2\pi \int_1^{65} \sqrt{u} \frac{du}{8}$$

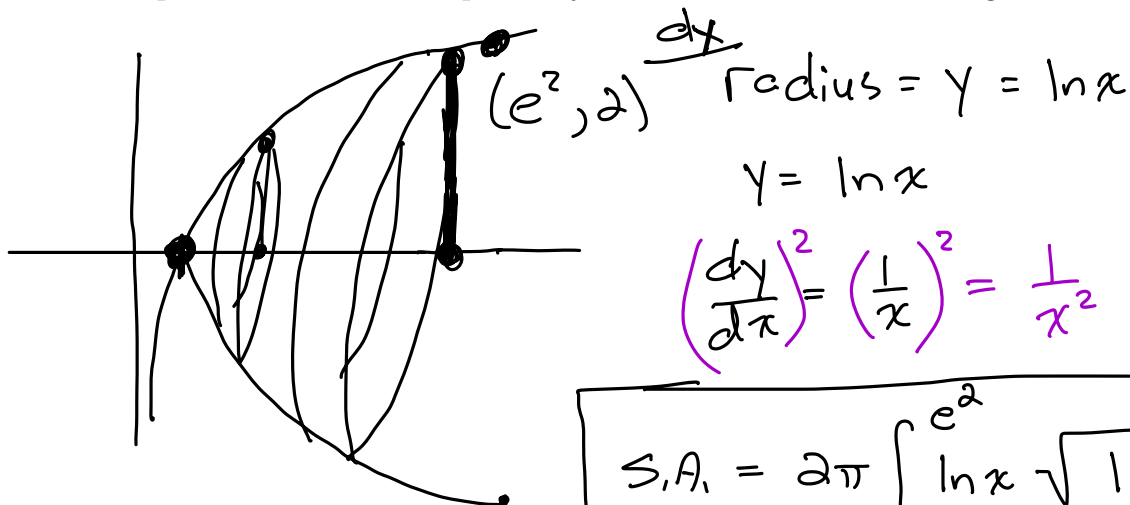
$$\text{Top: } 1 + 4(4)^2$$

$$\text{bottom: } 1 + 4(0)^2$$

$$= \frac{\pi}{4} \frac{2}{3} u^{3/2} \Big|_1^{65} = \frac{\pi}{6} u^{3/2} \Big|_1^{65} = \boxed{\frac{\pi}{6} (65^{3/2} - 1)}$$

0.0.4 Example:

Rotate the curve $y = \ln x$, $1 \leq x \leq e^2$ around the x -axis and find the surface area. Set it up with respect to x and with respect to y , and evaluate whichever integral looks easier.



$$y = \ln x$$

$$\left(\frac{dy}{dx}\right)^2 = \left(\frac{1}{x}\right)^2 = \frac{1}{x^2}$$

$$dS = \sqrt{1 + \frac{1}{x^2}} dx$$

$$S.A. = 2\pi \int_1^{e^2} \ln x \sqrt{1 + \frac{1}{x^2}} dx$$

$$\frac{dy}{e} \quad y = \ln x$$

$$\text{radius} = y$$

$$e^y = x$$

$$x = e^y$$

$$\left(\frac{dx}{dy}\right)^2 = (e^y)^2 = e^{2y}$$

$$dS = \sqrt{1 + e^{2y}} dy$$

$$S.A. = 2\pi \int_0^2 y \sqrt{1 + e^{2y}} dy$$

10.1 Parametric Curves

This section is an introduction to curves defined by parametric equations. We will graph them and examine their properties in 10.1. We will do calculus on parametric curves in section 10.2.

1 Parametric Curves

- Usually we define a curve by making y a function of x .
- We can also define both x and y as a function of a third variable t , which we often think of as time. The variable t is called a **parameter**.

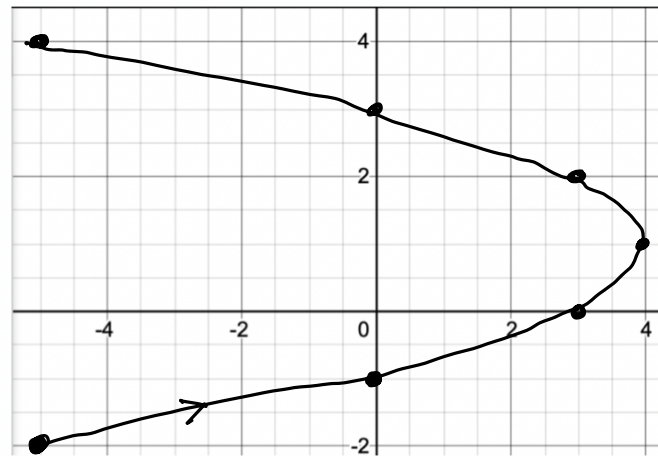
$$x = f(t)$$

$$y = g(t)$$

1.0.1 Example:

Plot the curve $x = 4 - t^2$ $y = t + 1$ from $t = -3$ to $t = 3$

t	x	y
-3	-5	-2
-2	0	-1
-1	3	0
0	4	1
1	3	2
2	0	3
3	-5	4



If we consider the graph to be the path of a particle, we also can observe its **speed** and **direction**.

It's also possible to remove the parameter by solving one equation for t and substituting into the other:

$$x = 4 - t^2 \quad y = t + 1$$

$$y - 1 = t$$

$$\boxed{x = 4 - (y - 1)^2} \leftarrow \text{sideways parabola}$$